Julien Martinelli

Research Fellow, Finnish Center for Artificial Intelligence, Aalto University

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Appointments

Apr 2025-present	**Research Fellow** Finnish Center for Artificial Intelligence, Aalto University, Finland
Mar 2023-Mar 2025	**Postdoctoral researcher** Inria Bordeaux Sud-Ouest, Bordeaux Population Health, France
Feb 2022-Feb 2024	**Postdoctoral researcher** Department of Computer Science, Aalto University, Finland
Oct-Nov 2023	**Visiting researcher** Inria Bordeaux Sud-Ouest, Bordeaux Population Health, France
Jul-Aug 2021	**Summer intern** Centre International de Rencontres Mathématiques, France

Education

February 2022	**PhD** École polytechnique, Inria Saclay, Institut Curie Saint-Cloud, France Thesis: On learning mechanistic models from time series data with applications to personalised chronotherapies
September 2018	**Master of Science in Applied Mathematics** Université Paris 7 (now Université Paris-Cité), France Thesis: mechanistic model learning from time-series data
May 2015	**Bachelor of Mathematics** Université Paris 5 (now Université Paris-Cité), France

Supervision & Leadership

PhD advising with Prof. Harri Lähdesmäki and Prof. Samuel Kaski at Aalto University

04/2026–present	**Tuan Vu**
09/2025–present	**Nicolas Blumer**, *Preliminary title: Sequential Decision-making with Expert feedback.*
01/2025–present	**Xinyu Zhang**, *In-context Black-Box Optimization.*
06/2023–present	**Marshal Sinaga**, *Robustness and Human-Centric Aspects in Bayesian Optimization.*

Other supervision (M) Master thesis (R) Research assistant (I) 2026 Summer intern (B) Bachelor thesis

06/2026–09/2026
(I) Mahtab Hosseinidolama, Mechanistic interpretability for tabular foundation models.
(I) Tomas Remis, Gaussian processes for biological dynamical systems.
(I) Quan Tran, High dimensional Bayesian optimization.

01/2026–present
(M) Barnik Roy Brata, In-context black-box optimization.

09/2025–03/2026
(M) Tuan Vu, Time-varying latent space Bayesian optimization.

06/2023-09/2024
(M) Xinyu Zhang, Preferential Amortized Black-Box Optimization.
Also (R) (2023) Probabilistic Neural Additive Models.

06/2023–09/2023
(B) Duong Le, Benchmarking Bayesian experimental design for drug design.

06/2023–09/2023
(B) Kaul Rajat, Learning biological ODE models from time-series data.

Teaching & Academic Service

Teaching

Fall 2025 **Modeling Biological Networks D**, Aalto University (MSc. level)
Gave 3 hours of lectures on data-driven inference of mechanistic ODEs and created the material myself.

Involved as a teaching assistant in exercise sessions:

- **Advanced Probabilistic Methods**, Aalto University (MSc. level): Fall semester 2022
- **Real Analysis**, Université Paris Cité (BSc. level): Spring semester 2020 and 2021
- **Mathematics and Calculus**, Université Paris Cité (BSc. level): Spring semester 2019

Academic service

Reviewer: NeurIPS, ICML, ICLR, AISTATS, UAI (annually since 2024); AAAI, JMLR, PAMI, TCBB, ECCB, CMSB.
Membership: European Laboratory for Learning and Intelligent Systems Society (ELLIS).
Education and workforce development: Contributor to NSF Thrust 4 (EWD), supporting training material on biofoundry workflows, and AI-enabled automation.

Invited talks

- European Conference on Mathematical and Theoretical Biology, Graz (2026), mini symposium “Automated discovery of dynamical digital twins from time series”
- European Conference on Mathematical and Theoretical Biology, Toledo (2024), mini symposium “Gaussian processes and inference for dynamical systems”
- Biostatistics Seminar, Bordeaux Population Health, France (2023)
- Public Seminar, CRiStAL team, Université de Lille, France (2023)
- AI Day, Finland (2022)
- Workshop on Hybrid models and methods in systems medicine, Institut Curie, Paris (2022)

Publication list

Publications

X. Zhang, C. Hassan, **J. M.**, D. Huang, and S. Kaski. In-Context Multi-Objective Optimization. In *International Conference on Learning Representations*, 2026. link

C. Métayer, A. Ballesta, and **J. M.** Data-driven Discovery of Digital Twins in Biomedical Research. *Briefings in Bioinformatics*, 2026. link

J. M. Position: Biology is the Challenge Physics-Informed ML needs to Evolve. In *Advances in Neural Information Processing Systems*, 2025. **6% acceptance rate**. link

M. Sinaga, **J. M.**, and S. Kaski. Robust and Computation-Aware Gaussian Processes. In *Advances in Neural Information Processing Systems*, 2025. link

S. Sloman, **J. M.**, and S. Kaski. Proxy-informed Bayesian Transfer Learning with Unknown Sources. In *Conference on Uncertainty in Artificial Intelligence*, 2025. link

X. Zhang, D. Huang, S. Kaski, and **J. M.** PABBO: Preferential Amortized Black-Box Optimization. In *International Conference on Learning Representations*, 2025. **Spotlight (top 5%)**. link

Y. Nahal, J. Menke, **J. M.**, M. Heinonen, M. Kabeshov, J. J. Paul, E. Nittinger, O. Engkvist, and S. Kaski. Human-in-the-loop Active Learning for Goal-Oriented Molecule Generation. *Journal of Cheminformatics*, 2024. link

S. Sloman, A. Bharti, **J. M.**, and S. Kaski. Bayesian Active Learning in the Presence of Nuisance Parameters. In *Conference on Uncertainty in Artificial Intelligence*, 2024. **Oral**. link

J. M., A. Bharti, A. Tiihonen, S. T. John, L. Filstroff, S. Sloman, P. Rinke, and S. Kaski. Learning Relevant Contextual Variables within Bayesian Optimization. In *Conference on Uncertainty in Artificial Intelligence*, 2024. link

C. Frioux, S. Huet, S. Labarthe, **J. M.**, T. Malou, D. Sherman, M.-L. Taupin, and P. Ugalde-Salas. Accelerating Metabolic Models Evaluation with Statistical Metamodels: Application to Salmonella Infection Models. *ESAIM: Proceedings and Surveys*, 2023. link

P. Mikkola, **J. M.**, L. Filstroff, and S. Kaski. Multi-Fidelity Bayesian Optimization with Unreliable Information Sources. In *International Conference on Artificial Intelligence and Statistics*, 2023. link

J. Hesse*, **J. M.***, O. Aboumanify, A. Ballesta, and A. Relógio. A Mathematical Model of the Circadian Clock and Drug Pharmacology to Optimize Irinotecan Administration Timing in Colorectal Cancer. *Computational and Structural Biotechnology Journal*, 2021. Co-first author. link

J. M., S. Dulong, X.-M. Li, M. Teboul, S. Soliman, F. Lévi, F. Fages, and A. Ballesta. Model Learning to Identify Systemic Regulators of the Peripheral Circadian Clock. *Bioinformatics*, 2021. link

J. M., J. Grignard, S. Soliman, and F. Fages. On Inferring Reactions from Data Time Series by a Statistical Learning Greedy Heuristics. In *International Conference on Computational Methods in Systems Biology*, 2019a. link

Preprints

J. M., M. Sinelnikov, H. Lähdesmäki, Q. Clairon, and M. Prague. Bayesian Nonparametric Mixed-Effect ODEs with Gaussian Processes, 2026b. arXiv. link

M. Sinaga*, **J. M.***, T. Turpeinen, and S. Kaski. Online Sharp-Calibrated Bayesian Optimization, 2026. arXiv. link

N. Blumer, **J. M.**, and S. Kaski. In-Context Black-Box Optimization with Unreliable Feedback, 2026. arXiv. link

J. M., I. Rebai, D. W. Haas, and J. Bertrand. Joint Bayesian Inference of Genetic Effect Sizes and PK Parameters in Nonlinear Mixed-Effects Models, 2026a. arXiv. link

T. A. Vu, **J. M.**, and H. Lähdesmäki. Time-Aware Latent Space Bayesian Optimization, 2026. arXiv. link

M. Sinaga, **J. M.**, V. Garg, and S. Kaski. Anchor-Based Heteroscedastic Noise for Preferential Bayesian Optimization, 2024a. arXiv. link

J. M., J. Grignard, S. Soliman, A. Ballesta, and F. Fages. Reactmine: a Statistical Search Algorithm for Inferring Chemical Reactions from Time Series Data, 2022. arXiv. link

Workshop communications

M. Sinaga, **J. M.**, and S. Kaski. Computation-Aware Robust Gaussian Processes. In *NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty*, 2024b. link

X. Zhang, **J. M.**, and S. T. John. Challenges in Interpretability of Additive Models. In *IJCAI 2024 Workshop on Explainable Artificial Intelligence*, 2024. link

J. M., Y. Nahal, D. Lê, O. Engkvist, and S. Kaski. Leveraging Expert Feedback to Align Proxy and Ground Truth Rewards in Goal-Oriented Molecular Generation. In *NeurIPS 2023 Workshop on New Frontiers of AI for Drug Discovery and Development*, 2023b. link

J. M., A. Bharti, A. Tiihonen, L. Filstroff, S. T. John, S. Sloman, P. Rinke, and S. Kaski. Learning Relevant Contextual Variables within Bayesian Optimization. In *NeurIPS 2023 Workshop on Adaptive Experimental Design and Active Learning in the Real World*, 2023a. link

M. Sinaga, **J. M.**, and S. Kaski. Preferential Heteroscedastic Bayesian Optimization with Informative Noise Priors. In *NeurIPS 2023 Workshop on Adaptive Experimental Design and Active Learning in the Real World*, 2023. link

J. M., J. Grignard, S. Soliman, and F. Fages. A Statistical Unsupervised Learning Algorithm for Inferring Reaction Networks from Time Series Data. In *ICML 2019 Workshop on Computational Biology*, 2019b. link

PhD thesis

J. M. *On Learning Mechanistic Models from Time Series Data with Applications to Personalised Chronotherapies*. PhD thesis, École Polytechnique, 2022. link